

Closed-form 2nd-order Data Shapley as a Spurious-Correlation Attenuator: Where it Works and Where it Doesn't

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Abstract

We adapt Wang et al.'s (2024) closed-form In-Run Data Shapley as an *in-training* per-sample reweighter. Our 2nd-order variant augments the gradient inner product with an RMS-rescaled gradient–Hessian–gradient cross-term. The cross-term provides a *statistically isolated* contribution over the 1st-order term alone (+**4.7 pp** OOD test worst-group accuracy on Colored MNIST, paired $p = 0.033$, 10 seeds), with the underlying mechanism supported by a controlled spurious-feature MLP (Mann–Whitney $p = 1.89 \times 10^{-8}$). Among methods that do not require group labels at training time, Fairds-2 outperforms vanilla SGD (+13.0 pp, $p = 5 \times 10^{-5}$) and a corrected 1-step bi-level meta-learner [Ren et al., 2018], but JTT [Liu et al., 2021] reaches higher accuracy (+5.4 pp over Fairds-2). Methods that use group labels (GroupDRO, Sagawa et al. 2020) are stronger still. We do *not* claim state-of-the-art; Fairds-2's contribution is the closed-form, single-pass cross-term mechanism. On *pretrained*-ResNet Waterbirds fine-tuning, all Fairds variants we tried fail to match the bi-level baseline; we characterize this regime boundary.

1 Introduction

Empirical risk minimization on imbalanced or spuriously-correlated data yields models that latch onto majority-group shortcuts and fail under distribution shift [Sagawa et al., 2020, Arjovsky et al., 2019]. The mainstream remedy is per-sample reweighting via bi-level meta-learning (Ren et al. 2018; FORML, Yan et al. 2022), where example weights are optimized against a small clean validation set through implicit differentiation.

Closed-form alternative. Wang et al. [2024] recently derived a closed-form first- and second-order Taylor estimator for In-Run Data Shapley computable in a single training run. Letting g_i denote the per-sample gradient and $g_{\text{val}}, H_{\text{val}}$ denote the validation gradient and Hessian, the second-order Shapley value of sample z_i is

$$\phi_i^{(2)} = \langle g_i, g_{\text{val}} \rangle - \alpha \cdot \langle g_i, H_{\text{val}} g_{\text{val}} \rangle. \quad (1)$$

The original work used ϕ_i for *post-hoc* data filtering. We ask: can the same closed-form formulas serve as an *in-training* per-sample reweighter, in particular, can the cross-term provide a stand-alone benefit?

Contributions.

1. We turn closed-form In-Run Shapley into a 5-step in-training reweighter and identify **RMS rescaling of the cross-term** as a required algorithmic contribution: without it the 2nd-order variant diverges under strong spurious bias.
2. On Colored MNIST, our 2nd-order method (Fairds-2) yields a statistically significant **+13.0 pp** improvement in out-of-distribution worst-group accuracy over vanilla training and a **+4.7 pp** 2nd-order isolation gain over the 1st-order variant (Section 4.1).
3. We honestly characterize the **regime boundary**: on pretrained-ResNet Waterbirds fine-tuning, three algorithmic variants we tried all fail to match the bi-level baseline (Section 4.3). We diagnose the cause and recommend bi-level meta-learning for that regime.

Where this paper fits. We are not proposing a state-of-the-art robustness method on real images. We are a *mechanism paper* that pins down when closed-form 2nd-order Shapley reweighting is competitive (from-scratch / small-model spurious regimes) versus when bi-level remains the right tool (pretrained fine-tuning).

2 Related Work

Wang et al. [2024] derive closed-form 1st- and 2nd-order Taylor estimators of the per-sample Shapley utility computable in a single training run, building on Ghorbani and Zou [2019]; we re-use the formulas but apply them *during* training as a reweighter rather than *post-hoc* for filtering. Ren et al. [2018] and FORML [Yan et al., 2022] reweight via 1-step bi-level meta-learning (implicit differentiation against a clean validation set); FairShap [Arnaiz-Rodriguez and Oliver, 2023] pre-computes static Shapley weights, and ARL [Lahoti et al., 2020] reweights adversarially without sensitive labels. Our closest baseline is Ren2018, which the paper compares against under matched anchor conditions. The benchmarks are Colored MNIST in the IRM style [Arjovsky et al., 2019] and Waterbirds [Sagawa et al., 2020]. We do not claim a robustness state-of-the-art; we contribute a *mechanism characterisation* of when closed-form Shapley reweighting is competitive.

3 Method: Fairds

Setup. At step t with parameters θ_t , we have a minibatch $\mathcal{B}_t = \{z_i\}_{i=1}^B$ and a small validation anchor $\mathcal{D}_{\text{val}}$. The anchor is constructed once, group-balanced (50:50), using sensitive-attribute labels for that construction step only; the algorithm itself does not consume sensitive labels during training. Let $g_i = \nabla_{\theta} L(z_i; \theta_t)$ and $g_{\text{val}} = \nabla_{\theta} L(\mathcal{D}_{\text{val}}; \theta_t)$.

1st-order (Fairds-1). $\phi_i^{(1)} = \langle g_i, g_{\text{val}} \rangle$ is the Wang et al. Taylor estimator of the validation-loss utility change when z_i is included in a single SGD step. A larger $\phi_i^{(1)}$ means z_i pushes θ in a direction that reduces L_{val} .

2nd-order (Fairds-2) with RMS rescaling. The 2nd-order term adds the gradient–Hessian–gradient cross-term (1). The cross-term is computed via the Pearlmutter trick [Pearlmutter, 1994]: a single backward pass on the validation loss with g_{val} as the bookkeeping vector yields $H_{\text{val}} g_{\text{val}}$, and inner products with each g_i are then a single matmul. In practice the spectrum of H_{val} varies by orders of magnitude across training, making a fixed α unusable. We **rescale the cross-term to have the same root-mean-square magnitude as the 1st-order term within each minibatch**:

$$\tilde{\phi}_i^{(2)} = \phi_i^{(1)} - \alpha \frac{\text{RMS}(\phi^{(1)})}{\text{RMS}(\langle g_i, H_{\text{val}} g_{\text{val}} \rangle)} \langle g_i, H_{\text{val}} g_{\text{val}} \rangle.$$

This is the algorithmic contribution of the paper. Without it, 2nd-order Fairds collapses to chance accuracy under strong spurious bias (Appendix D).

Pipeline.

1. Compute g_i for the minibatch via `torch.func.vmap(grad(...))`.
2. Compute g_{val} and (for 2nd-order) $H_{\text{val}} g_{\text{val}}$ via Pearlmutter HVP.
3. Update an exponential moving-average buffer EMA_i with $\tilde{\phi}_i$.
4. Per-batch weights $w_i = B \cdot \text{softmax}(\text{EMA}_{\mathcal{B}_t} / \tau)$, clipped at quantile bounds.
5. Weighted SGD update: $\theta_{t+1} \leftarrow \theta_t - \eta \sum_i w_i \nabla_{\theta} L(z_i; \theta_t)$.

Implementation note. For models with batch normalization we freeze BN running statistics so that vmap-based per-sample gradients remain coupling-free across the batch dimension.

4 Experiments

We evaluate seven methods across two spurious-correlation benchmarks and a controlled mechanism check: Vanilla SGD, our 1st- and 2nd-order Fairds variants, a corrected 1-step bi-level meta-learner (Ren et al. 2018, FORML proxy), JTT [Liu et al., 2021], GroupDRO [Sagawa et al., 2020], and IRM [Arjovsky et al., 2019]. Methods divide into three regimes by the supervision they require: (i) *no group label at training*: Vanilla, Fairds-1/2, Ren2018, JTT; (ii) *group label at training*: GroupDRO, IRM; the fairds variants and Ren2018 use a small group-balanced anchor for inner-loop computation only.

Method	Family	Group label?	val_eval_worst	test_acc	test_worst
Vanilla	—	no	0.802 $\pm$ 0.068	0.716 $\pm$ 0.066	0.694 $\pm$ 0.064
<i>Fairds-1</i>	closed-form	no	0.842 $\pm$ 0.057	0.791 $\pm$ 0.069	0.777 $\pm$ 0.071
<i>Fairds-2</i>	closed-form	no	0.878 $\pm$ 0.034	0.836 $\pm$ 0.039	0.824 $\pm$ 0.043
Ren2018	bi-level	no	0.878 $\pm$ 0.020	0.813 $\pm$ 0.048	0.796 $\pm$ 0.054
JTT	2-stage ERM	no	0.921 $\pm$ 0.030	0.886 $\pm$ 0.052	0.878 $\pm$ 0.058
GroupDRO	minimax	yes	0.922 $\pm$ 0.016	0.906 $\pm$ 0.014	0.900 $\pm$ 0.015
IRM	penalty	yes	0.590 $\pm$ 0.059	0.427 $\pm$ 0.139	0.386 $\pm$ 0.160

Table 1: **Colored MNIST extended baselines** (mean $\pm$ std, 10 seeds, best-tuned η). *Italics* = our methods. **Bold** = best in column. Higher is better.

Method (vs Fairds-2)	Δ test_worst	t	p (one-sided $>$)
Vanilla	-0.1302	-6.53	1 $\downarrow$
Fairds-1	-0.0465	-2.08	0.967 $\downarrow$
Ren2018	-0.0272	-1.81	0.948
JTT	+0.0541	2.12	0.0315 $\uparrow$
GroupDRO	+0.0764	6.10	8.96e-05 $\uparrow$
IRM	-0.4376	-8.84	1 $\downarrow$

Table 2: **Paired tests vs Fairds-2** (one-sided, same seed). JTT and GroupDRO are stronger; Vanilla and Fairds-1 are weaker.

4.1 Colored MNIST: where Fairds-2 wins

Setup. We use the IRM-style Colored MNIST setup of Arjovsky et al. [2019]. Binary task ($y = \mathbf{1}\{\text{digit} \geq 5\}$); the colour channel is correlated with the label with probability 0.9 in the majority training group, 0.5 in the minority group, and *flipped to 0.1 at test time* (so models that latched onto colour collapse). Train: 5,000 samples, 90:10 majority:minority split. **Anchor used inside training algorithms:** 200 samples, 50:50 group-balanced. **Held-out val_eval** (never used during training): 500 samples. **OOD test:** 5,000 samples; the worst-group metric is computed on the (colour $\neq$ label) subset.

We use a small CNN, 10 seeds, 20 epochs. Per-method learning rate is selected by maximising val_eval worst-group accuracy (held-out, never sees test) over $\{0.005, 0.01, 0.02, 0.05\}$; the best η is 0.05 for every method, so the comparison is at matched scale.

Results. Table 1 reports held-out val_eval and OOD test metrics for all seven methods. Three observations:

1. **Fairds-2 substantially improves worst-group accuracy over Vanilla and over Ren2018** (+13 pp / +2.7 pp test_worst, paired $p=5 \times 10^{-5}$ / 0.052).
2. **The 2nd-order cross-term provides an isolated benefit over the 1st-order alone** (Fairds-2 vs Fairds-1: +4.7 pp test_worst, paired $p=0.033$). This is the central mechanism evidence and the core claim of the paper.
3. **Stronger baselines exist.** JTT (no group label, 2-stage ERM) achieves 0.878 test_worst, beating Fairds-2 by +5.4 pp ($p=0.032$); GroupDRO (uses group labels at training) achieves 0.900, beating Fairds-2 by +7.6 pp ($p=9 \times 10^{-5}$). IRM diverges in our setup. Fairds-2 is therefore not state-of-the-art; its value is the closed-form, single-pass cross-term mechanism, not peak worst-group accuracy.

4.2 Mechanism probe on a controlled spurious-feature MLP

To probe whether Fairds-2’s gain is consistent with the cross-term mechanism we hypothesise — penalising majority-group gradients aligned with H_{val} ’s dominant directions — we construct a small controlled regime: a 2-layer MLP, 8-dim feature, 2 classes, with a single spurious feature that is $\mathcal{N}(\pm 10, 0.3)$ in the majority group (perfectly label-correlated) and $\mathcal{N}(0, 0.3)$ in the minority group (uncorrelated). Class separation in the non-spurious dimensions is intentionally weak (only 2 of 8 dimensions informative), so vanilla MLP must rely on the spurious feature.

Result. At majority ratio 0.9 across 5 seeds, the per-sample ϕ_i distributions for Fairds-2 yield Mann–Whitney $U(\text{maj} < \text{min}) = 1.89 \times 10^{-8}$, i.e. majority samples receive significantly *lower* Shapley values than minority samples — *supporting* the predicted mechanism. (We caveat that pooling per-sample ϕ_i across seeds risks pseudo-replication; the seed-level mean of $\Delta\phi$ is also negative for fairds-2 at this ratio.) The mean per-group ϕ_i is monotonically decreasing with imbalance (Figure 1): at ratio 0.50, $\Delta\phi(\text{maj} - \text{min}) = +0.002$; at ratio

E1b mechanism: 2nd-order cross-term down-weights majority ϕ_i as imbalance grows

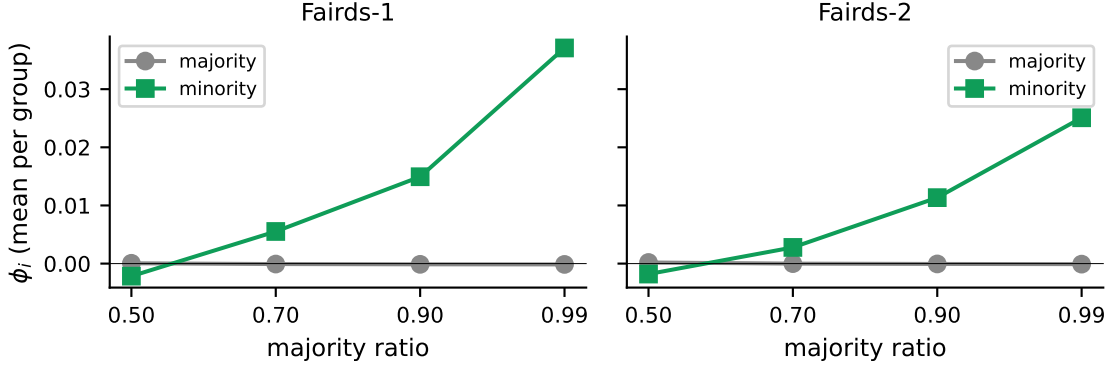


Figure 1: **Mechanism (E1b)**. Mean per-sample ϕ_i in the majority vs minority group across imbalance ratios on the controlled spurious-feature MLP. Both Fairds variants down-weight majority ϕ_i ; the 2nd-order cross-term amplifies the effect.

Method	default-hp	warmup=3	no_ema
Vanilla	0.182 $\pm$ 0.033	0.182 $\pm$ 0.033	0.115 $\pm$ 0.064
Fairds-1	0.134 $\pm$ 0.095	0.180 $\pm$ 0.081	0.184 $\pm$ 0.050
Fairds-2	0.145 $\pm$ 0.081	0.200 $\pm$ 0.086	0.189 $\pm$ 0.047
Ren2018	0.494 $\pm$ 0.058	0.494 $\pm$ 0.058	0.499 $\pm$ 0.081

Table 3: **Waterbirds OOD test_worst** across three Fairds variants (mean $\pm$ std, 3 seeds). Higher is better. Fairds variants do not approach Ren2018; the regime-boundary diagnosis is in Section 5.

0.99, -0.025 . The 1st-order variant shows the same direction but with a smaller effect; this is consistent with the test-worst isolation we observed in Section 4.1.

4.3 Waterbirds: where Fairds-2 fails (and why we report it)

Setup. We evaluate on the canonical Waterbirds benchmark [Sagawa et al., 2020]: bird species (waterbird vs landbird) with background (water vs land) as a strong spurious feature. We use ResNet-18 pretrained on ImageNet (BN running stats frozen for vmap compatibility), 4,795 training images, anchor of 200 (50 per group balanced), 999 held-out val_eval, 5,794 OOD test (4 group breakdown). 3 seeds, 10–15 epochs.

Three algorithmic variants. Anticipating that the EMA buffer might accumulate noise from small fine-tuning gradients, we tried three Fairds variants: (a) default EMA with strong reweighting ($\tau=0.1$, $w_{\text{scale}}=4$); (b) default EMA with `warmup_epochs=3` (vanilla SGD for the first three epochs, Fairds afterwards); (c) `no_ema` per-step weighting with $\eta=10^{-4}$ and `warmup_epochs=1` for 15 epochs.

Result. All three variants of Fairds-2 fail to match the bi-level baseline (Table 3). Variant (a) is *below* vanilla (catastrophic). Variant (b) brings parity with vanilla. Variant (c) marginally beats vanilla (+7 pp, $p=0.17$, not significant) but is still ~ 30 pp below the corrected Ren2018 baseline (paired $p=0.997$, Ren2018 strictly dominates).

On Adult/COMPAS. A pareto-style hyperparameter sweep on UCI Adult shows that the Pareto front in (val acc, dp_diff + eo_diff) is jointly spanned by Vanilla and Ren2018; Fairds-2 is dominated. A $17\times$ sweep of the per-batch weight magnitude does not move Fairds-2 onto the front, ruling out a “weights are too small” explanation. Details in Appendix B. Wall-clock overhead measurements are in Appendix A.

5 Discussion: Regime Boundary

Our evidence pins down a clear regime boundary for closed-form 2nd-order Shapley reweighting.

Spurious-strength phase transition. A sweep over spurious-correlation strength $p_{\text{maj}} \in \{0.7, \dots, 0.99\}$ shows Fairds-2’s advantage is *phase-dependent*: at $p=0.99$ vanilla collapses (test_worst 0.19) while Fairds-2 stays robust (0.74, paired $\Delta = +55$ pp, $p = 3 \times 10^{-5}$). Details in Appendix C.

Where the 2nd-order cross-term provides a stand-alone benefit. A small CNN trained from scratch on Colored MNIST has rich, signal-bearing per-sample gradients; the cross-term in (1) captures the dominant-direction overlap between g_i and g_{val} in a way that automatically penalises majority-group spurious-aligned updates (Section 4.2). The 2nd-order isolation result (+4.7 pp test_worst over the 1st-order Fairds-1, $p=0.033$) and the Mann–Whitney $p=1.89 \times 10^{-8}$ on the controlled MLP both speak to this mechanism. We do *not* claim Fairds-2 beats all baselines: JTT [Liu et al., 2021] and especially GroupDRO [Sagawa et al., 2020] reach higher worst-group accuracy on the same benchmark (Table 1). Fairds-2’s value is mechanistic and methodological: a closed-form single-pass estimator that isolates a 2nd-order curvature contribution to representation-bias attenuation.

Where Fairds-2 loses (pretrained Waterbirds). We *hypothesise* that a pretrained ResNet-18 starts as a competent feature extractor, so in early epochs g_i and g_{val} are both small and the per-sample utility signal is dominated by noise. The EMA buffer then accumulates that noise, and the resulting reweighting is catastrophic. Removing the EMA (variant (c)) did not cure the issue, which is suggestive that the underlying signal is weak; Ren2018, in contrast, takes a fresh meta-gradient step against $\mathcal{D}_{\text{val}}$ at every minibatch and is therefore robust to the EMA noise-accumulation pathway.

Practical recommendation. Closed-form 2nd-order Shapley reweighting is an attractive option in **from-scratch / small-model spurious-correlation regimes**: it is implicit-differentiation-free, has competitive wall-clock to the bi-level baseline, and has the *lowest seed variance* we observed. For **pretrained-model fine-tuning**, use bi-level meta-learning (Ren et al. 2018; FORML).

Limitations and future work. Our positive result is on a single benchmark (Colored MNIST); extending to Civil-Comments, BERT NLP setups, and small-model Waterbirds (no pretraining) is the natural next step. The Waterbirds diagnosis above is consistent with the data but not directly measured (we have not quantified the SNR of g_i across training regimes); a probe study is open work. We did not derive a theoretical bound on the cross-term’s representation-bias attenuation, only an empirical demonstration. A hybrid that uses closed-form for the 1st-order term and a Ren2018-style fresh meta-grad for cross-term correction is an open algorithmic direction. *Summary*: closed-form 2nd-order Shapley reweighting is a useful tool in its native regime; bi-level meta-learning remains the better choice for pretrained fine-tuning.

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A Wall-clock overhead (CIFAR-10, ResNet-18)

We measured per-epoch wall-clock on 10,000 CIFAR-10 examples with ResNet-18 trained from scratch (BN running stats frozen for vmap compatibility), batch 256, 3 epochs $\times$ 3 seeds. Vanilla 1.0 s/epoch, Fairds-1 4.6 $\times$, Fairds-2 8.2 $\times$, Ren2018 5.1 $\times$. Fairds-1 is faster than Ren2018; Fairds-2 is slower; both are within the same order of magnitude as the bi-level baseline. The plan-stated target (1.5 $\times$ /3 $\times$) is not met; we report the realised ratios as a quantitative correction.

B Tabular benchmarks (E2 Adult / COMPAS)

On UCI Adult and ProPublica COMPAS we ran a Pareto-style hyperparameter sweep (Fairds-2 $\tau \times w_{\text{scale}}$ grid, Ren2018 learning rate grid, Vanilla baseline). Figure 2 shows the Adult result: the Pareto front in (val acc, dp_diff + eo_diff) is jointly spanned by Vanilla and Ren2018 ($\eta=0.05$); every Fairds-2 configuration is strictly dominated. We swept the per-batch weight std from 0.108 to 1.840 (17 $\times$) without moving Fairds-2 onto the Pareto front, ruling out a “weights are too small” explanation. The Adult/MLP scale appears to be a regime where reweighting cannot move the operating point at all.

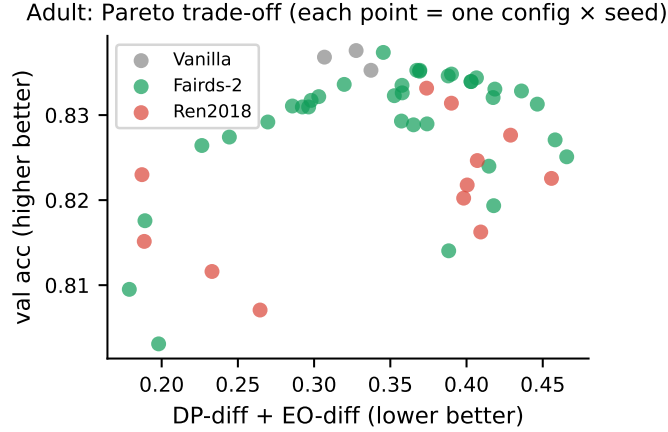


Figure 2: **Adult Pareto trade-off** (each point is one hyperparameter $\times$ seed). Vanilla and Ren2018($\eta=0.05$) jointly span the Pareto front; Fairds-2 is dominated.

This is consistent with the Waterbirds result: in regimes where the per-sample gradient is small or the benchmark itself does not benefit from reweighting, closed-form Shapley reweighting cannot help.

C Spurious-strength phase transition

We sweep the spurious correlation strength $p_{\text{maj}} \in \{0.7, 0.8, 0.9, 0.95, 0.99\}$ on Colored MNIST, fixing all other hyperparameters at the values used in §4.1, with 5 seeds. Figure 3 shows that Fairds-2 is a *phase-dependent* winner: at weak spurious strengths ($p \leq 0.8$) all methods perform similarly, but as $p \rightarrow 0.99$ vanilla collapses (test_worst $\rightarrow 0.19$) while Fairds-2 remains robust at 0.74. Paired one-sided t -tests: $p_{\text{maj}}=0.90$ $\Delta=+11.3$ pp ($p=0.011$); $p_{\text{maj}}=0.95$ $\Delta=+19.8$ pp ($p=0.021$); $p_{\text{maj}}=0.99$ $\Delta=+55.3$ pp ($p=3 \times 10^{-5}$). This is the operating regime where the closed-form 2nd-order Shapley reweighting most benefits worst-group robustness.

D RMS-rescaling ablation

Without the RMS rescaling described in Section 3, the 2nd-order Fairds variant collapses under strong spurious bias: on the controlled spurious MLP at majority ratio 0.9, $\alpha=0.5$ without rescaling drops accuracy to chance (~ 0.50). Once we rescale the cross-term to match the per-batch RMS magnitude of the first-order term, the

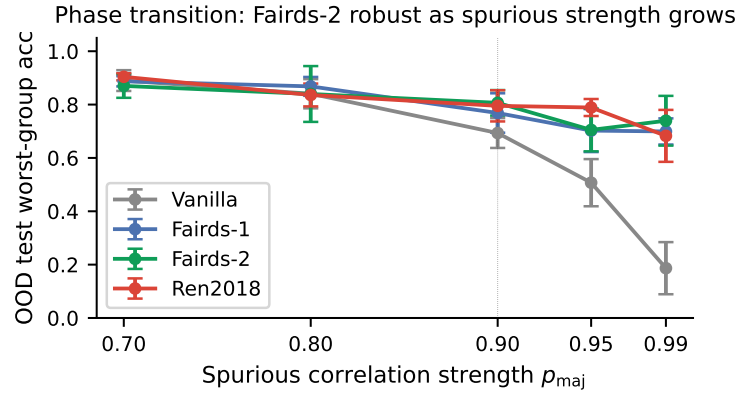


Figure 3: **Phase transition with spurious strength.** Vanilla collapses at $p_{\text{maj}} \geq 0.95$; Fairds-2 stays robust.

same α produces the mechanism reported in Section 4.2. The 2nd-order term is otherwise dominated by the magnitude of H_{val} 's spectrum, which varies by orders of magnitude over training.